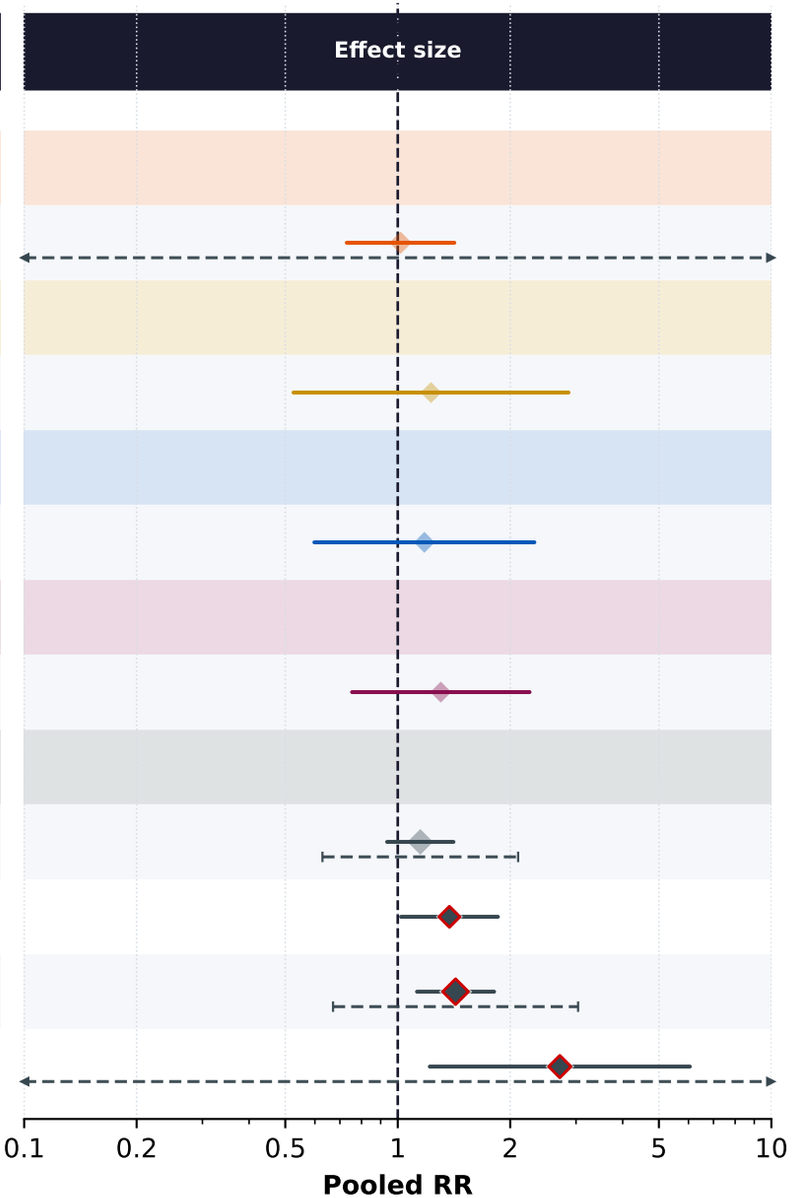
Exposures positively associated with renal cell carcinoma risk
Meta-analysis of dietary-intake studies

Exposure	Estimates	N	Cases	Pooled RR (95% CI)
CAROTENOIDS				
Lycopene	3	873,444	2,860	*1.02 (0.73-1.42)*
VITAMINS A, C, D, E, K				
Vitamin A	2	4,597	1,909	*1.23 (0.53-2.87)*
B VITAMINS				
Folic Acid	2	4,597	1,909	*1.18 (0.60-2.32)*
METABOLITES & AMINO ACIDS				
Dietary Protein Intake	2	776,349	2,168	*1.30 (0.75-2.25)*
OTHER				
Processed Meat Intake	6	1,766,660	5,212	*1.15 (0.94-1.41)*
Ultra Processed Food Intake	2	647,537	594	*1.38 (1.02-1.85)*
Red Meat	6	1,768,446	5,800	*1.43 (1.13-1.81)*
Dairy	3	3,506	1,009	*2.72 (1.22-6.05)*



Exposure group

Carotenoids · Vitamins A, C, D, E, K · B Vitamins · Metabolites & Amino Acids · Other

RR > 1 = positively associated with renal cell carcinoma risk. | ◆ pooled RR (size reflects number of estimates) | ◇ red outline = statistically significant (95% CI excludes 1.0) | ◆ faded = not statistically significant | dashed line below CI = 95% prediction interval (where available) | N and case totals sum estimates and may overlap across strata | arrowhead = interval continues beyond the axis

Include subexposures: No